An algorithm-independent lower limit on combinatorial fake tracks in high-occupancy tracking detectors

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Abstract: When the density of noise hits in a tracking detector becomes large compared with its resolution, some random combinations of hits are indistinguishable from real tracks: they pass the same goodness-of-fit cut that real tracks pass, and no reconstruction algorithm that keeps the efficiency for real tracks can reject them. We give their expected number, which is therefore a lower limit on the fake-track rate of any algorithm. We first treat a few parallel planes with uniform noise and straight tracks. The expected number is the product of the numbers of noise hits and the probability that one random combination passes the cut, and that probability, which can be as small as 10^{-23}, follows from the geometry: it is the number of distinguishable tracks in the acceptance, times the volume of a χ^2 sphere, divided by the volume of the space of hit positions, all in units of the resolution. We then give the general solution, for layers of any shape and orientation (cylinders, tilted planes, disks), helical tracks in a solenoidal field, noise densities that vary across the detector, and tracks required to have hits on at least k of N layers. The results are checked against direct Monte Carlo and brute-force counting in small detectors.

Keywords: Pattern recognition, cluster finding, calibration and fitting methods; Performance of High Energy Physics Detectors; Analysis and statistical methods

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Contents

1 Introduction **2**

2 The simple case: parallel planes, uniform noise, straight tracks **3**
2.1 Detector, noise and acceptance . . . 3
2.2 The counting identity and the lower limit . . . 3
2.3 The probability that one combination passes . . . 4
2.4 Resolution scaling and exact rescaling . . . 5
2.5 Requiring at least k of N planes . . . 5
2.6 A numerical example . . . 8

3 The general solution **8**
3.1 Arbitrary surfaces and local densities . . . 8
3.2 Helical tracks in a solenoidal field . . . 9
3.3 At least k of N layers in a general detector . . . 9
3.4 Evaluating the integral . . . 9

4 Validation **10**

5 Uncertainties and limitations **11**

6 Conclusions **13**

A Notation **14**

B The volume of the tube **14**

C Constants **14**

D Code and data availability **14**

1 Introduction

Track reconstruction in a high-occupancy environment must find the few hits left by charged particles among many unrelated ones. The problem is familiar at hadron colliders, where pile-up fills the inner layers [1, 2], and it is extreme at a muon collider, where the decay products of the circulating muons produce a beam-induced background of millions of hits per bunch crossing in the tracker [3, 4, 5, 6, 7]. Some combinations of background hits happen to line up well enough to be accepted as tracks. The resulting fake-track rate is normally measured by running a specific reconstruction algorithm on fully simulated events [8, 9, 10], so the number obtained belongs to that algorithm.

This paper asks a different question: at what hit density does track reconstruction become impossible for *any* algorithm? The answer exists because a combination of noise hits that fits a track hypothesis within the resolution carries no information distinguishing it from a real track. An algorithm that keeps the real tracks that pass a given goodness-of-fit requirement must also keep these combinations. Their expected number is therefore a lower limit on the fake rate of every algorithm with that efficiency, and it can be computed from the detector geometry, the resolution and the hit densities alone.

The expected number is the product of the numbers of noise hits and the probability P that one random combination passes the cut. The difficulty is that P is tiny, of order 10^{-23} for six layers of realistic size, far beyond direct simulation. Earlier work [11] (section 8.5) estimated such probabilities with a Monte Carlo at inflated resolution, scaled back by the product of the resolution ratios. Here we make that argument exact: P follows a power law in the cut, $P \simeq K c^{n_{\mathrm{dof}}/2}$, whose exponent is fixed by the number of degrees of freedom and whose constant K is a geometric quantity, the volume of the set of accepted tracks in the space of hit positions.

Related ideas appear in several fields. The probability that k random points are aligned within a strip of width w scales as $(w/L)^{k-2}$ [12, 13]; our result contains this law as a special case. The "number of false alarms" of a contrario detection in image analysis [14] is, like our estimate, an expected number of chance events that needs no independence assumption. The volume of a tube around a manifold, which gives our constant, was computed by Hotelling and Weyl [15, 16] and is used in statistics for significance tests in nonlinear models [17, 18]. Methods that make a rare event common by changing a scale and then extrapolate back are used in structural reliability [19, 20]; here the rescaling is an exact symmetry and nothing is extrapolated by a fit. In particle physics, accidental coincidences between counters [21] and fake roads in track triggers [22, 23, 24] are estimated with the same product-of-occupancies reasoning, but to our knowledge no exact general expression has been given for combinations accepted by a χ^2 cut.

Section 2 introduces the problem and the method on the simplest case: a few parallel planes, uniform noise and straight tracks, with numerical examples. Section 3 gives the general solution, for any detector surfaces, helical tracks in a solenoidal field and noise densities that vary across the detector. Section 4 checks the results against direct Monte Carlo and brute-force counting,

and section 5 discusses the limitations. An application to the tracker of a muon collider is the subject of a companion paper [ref].

2 The simple case: parallel planes, uniform noise, straight tracks

2.1 Detector, noise and acceptance

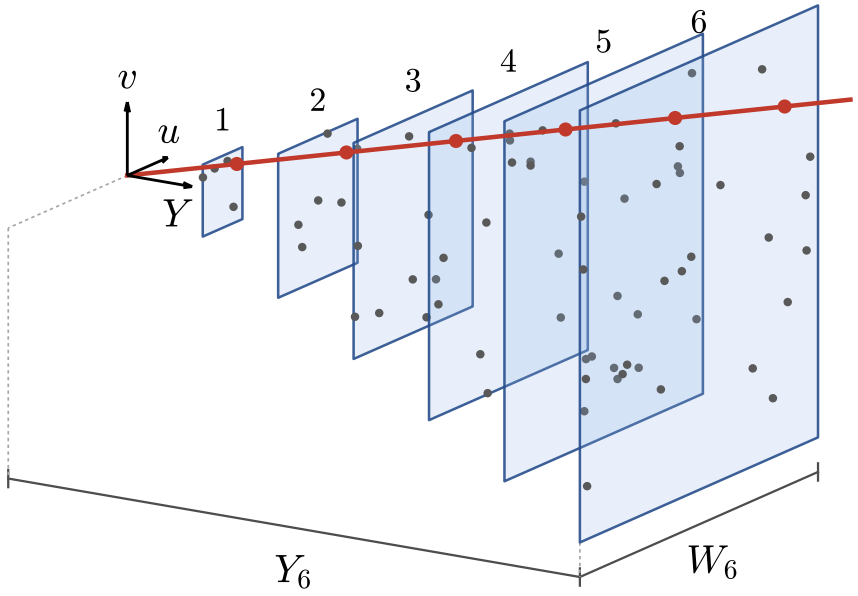


Figure 1: The simple case: six parallel square planes at depths Y_i, of side W_i (the projective tower of section 2.6), each with a few uniformly distributed noise hits (grey). A straight track (red) crosses every plane; each plane measures the two coordinates u and v of its hit.

The detector consists of N parallel planes at depths Y_i (figure 1). Plane i measures two coordinates u and v of each hit, with Gaussian resolutions $\sigma_{u,i}$ and $\sigma_{v,i}$, over a region of area A_i (in the examples a square of side W_i). Noise hits are distributed uniformly over each plane, independently of each other, with n_i hits on plane i (a fixed number, or a Poisson variable of mean n_i; the results are the same). Tracks are straight lines, $u = a_u + b_u Y$ and $v = a_v + b_v Y$.

A combination of one hit per plane is fitted by least squares to the track model, with $p = 4$ parameters; with $m = 2N$ measured coordinates the fit has $n_{\mathrm{dof}} = m - p$ degrees of freedom. The combination is accepted as a track if $\chi^2 < c$ and the fitted parameters lie in an acceptance region Θ. The cut c is set by the efficiency required for real tracks; we use a one-sided 3σ cut, $\Pr(\chi^2_{n_{\mathrm{dof}}} > c) = 1.35 \times 10^{-3}$, which gives $c = 17.80$, 21.74 and 25.36 for $n_{\mathrm{dof}} = 4$, 6 and 8.

2.2 The counting identity and the lower limit

Let P be the probability that one combination of independent, uniformly distributed hits is accepted. The expected number of accepted noise combinations is

$$E = P \prod_{i=1}^{N} n_i, \tag{1}$$

by linearity of expectation: E is the sum, over the $\prod_i n_i$ combinations, of the probability that each is accepted. The identity is exact for the mean even though combinations that share hits are correlated; correlations affect the variance of the number of fakes, not its mean.

A combination of noise hits that passes the cut is, as data, a valid track: some particle with parameters in Θ could have produced those hits with the observed residuals. Consider any algorithm that accepts every combination of hits compatible with a track in the acceptance, $\chi^2 < c$ and $\theta \in \Theta$, so that its efficiency for real tracks is at least that of the cut. Such an algorithm accepts all the noise combinations counted by eq. (1), and E is a lower limit on its number of fake tracks. An algorithm can produce fewer fakes, but only by also losing real tracks: by tightening c or Θ, which the same formula quantifies, or by rejecting candidates according to their surroundings, which removes real tracks in the same surroundings. When E approaches one per event, the information needed to reconstruct tracks with the stated efficiency is no longer in the data.

Strictly, E counts combinations, while an algorithm reports tracks: two accepted combinations that share all but one hit would be reported as one fake. The lower limit is therefore on the expected number of *distinct* fakes, which is smaller than E by a factor $e^{-\eta/2}$ set by the window occupancy η (section 2.5). Wherever tracking is possible $\eta \ll 1$ and the two coincide, so we use E for both where the difference does not matter.

The limit counts only combinations made entirely of noise hits. Combinations in which some hits belong to a real track and others are noise are also fakes in the usual sense; they only add to the number given here.

2.3 The probability that one combination passes

Whitened coordinates. Dividing every measured coordinate by its resolution turns the weighted fit into an unweighted one. A combination of hits becomes a point x in an m-dimensional space, uniformly distributed in a box B whose sides are $W_i/\sigma_{u,i}$ and $W_i/\sigma_{v,i}$, of volume $|B| = \prod_i A_i/(\sigma_{u,i}\sigma_{v,i})$. Each track θ produces an ideal point $f(\theta)$, and as θ runs over the accepted tracks whose hits lie inside every plane, these points form a p-dimensional surface M, the *track manifold*. The fit finds the point of M nearest to x, and

$$\chi^2(x) = \min_\theta |x - f(\theta)|^2 \tag{2}$$

is the squared distance from x to M. In these coordinates every plane is a box and every acceptance region a sphere, whatever the sizes and resolutions.

The small-χ^2 law. The accepted combinations are the points within distance $\sqrt{c}$ of M: a thin tube around the track manifold. In each of the n_{dof} directions perpendicular to M the tube is a sphere of radius $\sqrt{c}$, of volume $V_{n_{\mathrm{dof}}} c^{n_{\mathrm{dof}}/2}$ with $V_d = \pi^{d/2}/\Gamma(d/2+1)$. To leading order its volume is the area of M times this sphere volume, the result of Hotelling and Weyl [15, 16], and since x is uniform in B,

$$P \simeq K\, c^{n_{\mathrm{dof}}/2}, \qquad K = \frac{V_{n_{\mathrm{dof}}}\,\mathrm{Vol}(M)}{|B|}, \qquad \mathrm{Vol}(M) = \int_{\Theta_B} \sqrt{\det(J^{\mathsf{T}}J)}\; \mathrm{d}^p\theta, \tag{3}$$

where $J = \partial f/\partial\theta$ is the $m \times p$ matrix of derivatives of the whitened ideal hits with respect to the track parameters, and Θ_B is the set of parameters in Θ whose ideal hits all lie inside the planes (appendix B).

The exponent depends only on the number of degrees of freedom. The constant has a simple reading: $J^{\mathsf{T}}J$ is the inverse covariance matrix C^{-1} of the fitted parameters, so $\mathrm{Vol}(M) = \int \mathrm{d}^p\theta/\sqrt{\det C}$ is the number of *distinguishable tracks* in the acceptance, the volume of parameter space counted in units of the parameter errors. The expected number of fakes is this number of tracks times the probability that, for one of them, random hits fall inside its χ^2 sphere on every plane.

For straight lines the Jacobian is constant: $\det(J^{\mathsf{T}}J)$ is the product over the two views of $(N\sum_i Y_i^2 - (\sum_i Y_i)^2)/\sigma^4$, and Θ_B is, in each view, the polygon of (a, b) for which $|a + bY_i| \le W_i/2$

on every plane. The integral reduces to the area of that polygon, a one-dimensional integral. The familiar small-argument form of the χ^2 distribution, $\Pr(\chi^2_{n_{\rm dof}} < c) \simeq (c/2)^{n_{\rm dof}/2}/\Gamma(n_{\rm dof}/2+1)$ for Gaussian residuals [25], has the same power for the same geometric reason, with a different constant because noise hits are uniform rather than Gaussian around a track. For k points in one view, eq. (3) reproduces the $(w/L)^{k-2}$ law of chance alignments [12, 13].

Corrections. Equation (3) holds when the sphere is small compared with the planes in whitened units. The leading corrections, from the corners of the acceptance region and from the curvature of M, are of relative order c in units of the whitened plane size; for a linear track model and a flat boundary the first-order edge effects cancel (appendix B). In any realistic detector the planes are hundreds to thousands of resolutions wide and the corrections are negligible.

2.4 Resolution scaling and exact rescaling

At fixed geometry, $\mathrm{Vol}(M) \propto \sigma^{-p}$ and $|B| \propto \sigma^{-m}$, so at fixed cut

$$P \propto \sigma^{n_{\rm dof}}, \tag{4}$$

and $P \propto \sigma_u^{n_u}\sigma_v^{n_v}$ when the two views have different resolutions and $n_u + n_v = n_{\rm dof}$ degrees of freedom. Equivalently, shrinking every length of the detector by a factor λ at fixed resolution multiplies every residual by $1/\lambda$ and χ^2 by $1/\lambda^2$, so that the probability in the real detector is exactly that in the shrunken one at a rescaled cut,

$$P_{\rm true}(c) = P_{\rm shrunk}(c/\lambda^2). \tag{5}$$

We use this identity to test eq. (3) by direct Monte Carlo: a detector shrunk until its smallest plane is two resolutions wide makes accepted combinations common. We call it the reference detector and $c' = c/\lambda^2$ the cut mapped onto it.

2.5 Requiring at least k of N planes

Why sets of planes. A real detector is not fully efficient, and a track finder that required a hit on every plane would lose every track with a missing hit. Track finders therefore accept candidates with hits on *at least* k of the N planes, five of six for example. Any such algorithm must also accept the noise combinations that pass on a subset of the planes. We therefore extend the counting of section 2.2 from one set of planes to all the sets that the algorithm accepts.

Candidates. Let S be a set of planes with $|S| \geq k$. A *candidate on* S is a choice of one hit on each plane of S, nothing being required of the planes outside S. It is fitted to the track model with the planes of S only, with $n_{\rm dof}(S) = 2|S| - 4$ degrees of freedom, and it is *accepted* if

$$\chi^2_S < c_S \quad \text{and} \quad \hat{\theta}_S \in \Theta, \tag{6}$$

where $\hat{\theta}_S$ are the fitted parameters. Each set has its own cut c_S, chosen at the same efficiency for real tracks as for the full set. With our one-sided 3σ choice the cut depends only on $n_{\rm dof}(S)$: for five of six planes, $c = 25.36$ for the full set ($n_{\rm dof} = 8$) and $c = 21.74$ for each of the six five-plane sets ($n_{\rm dof} = 6$).

The expected number of accepted candidates. We define $E(\geq k)$ as the expected number of accepted candidates, summed over all sets of at least k planes. A candidate on S is accepted with probability P_S, the probability of section 2.3 computed for the planes of S alone (their depths, sizes and resolutions), and there are $\prod_{i\in S} n_i$ candidates on S. By linearity of expectation, as in eq. (1),

$$E(\geq k) \;=\; \sum_{S:\,|S|\geq k} E_S, \qquad E_S \;=\; P_S \prod_{i\in S} n_i. \tag{7}$$

For at least five of six planes the sum has seven terms:

$$E(\geq 5) \;=\; P_{123456}\, n_1 n_2 n_3 n_4 n_5 n_6 \;+\; \sum_{j=1}^{6} P_{\bar{j}} \prod_{i \neq j} n_i, \tag{8}$$

where $\bar{j}$ denotes the five planes other than j. Note that a candidate on five planes is counted whether or not the sixth plane happens to have a hit near the track; this is what an algorithm accepting five of six planes does.

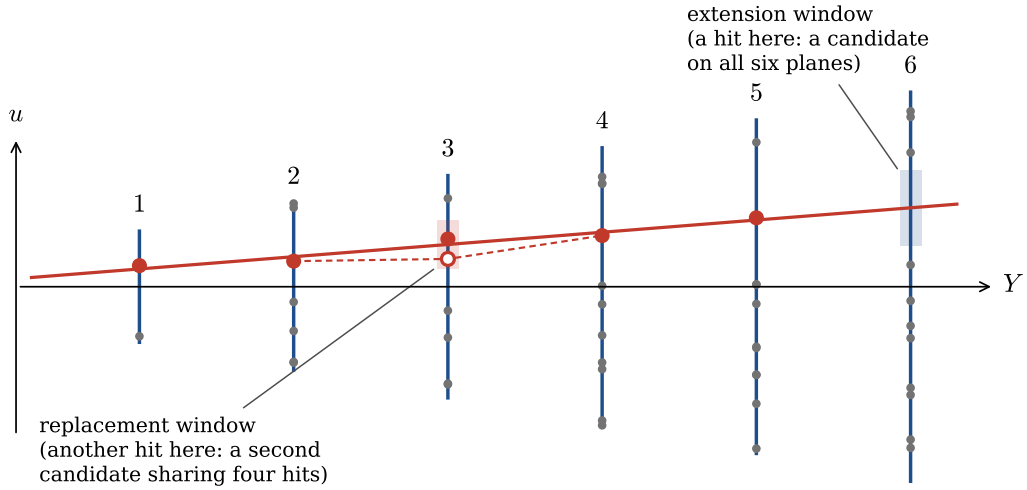


Figure 2: Candidates that share hits, shown in one view and not to scale. A candidate on planes 1–5 (filled red points) is accepted. The fit to planes 1, 2, 4 and 5 predicts where its hit on plane 3 must lie: any other hit in this *replacement window* (open point) forms a second accepted candidate on the same planes, sharing four hits with the first. The fit to planes 1–5 predicts plane 6: a hit in this *extension window* forms an accepted candidate on all six planes, which contains the first one.

Windows. Two accepted candidates can share hits, and the size of the overlap is governed by a single quantity, the number of hits expected in a *window* (figure 2). Take an accepted candidate on S and one of its planes, i. The fit to the other planes of S predicts the position $(\hat{u}_i, \hat{v}_i)$ of the track on plane i, with variances $V_{u,i}$ and $V_{v,i}$, and has its own $\chi^2_{S\setminus i}$. For a linear track model the χ^2 of the full set separates exactly into

$$\chi^2_S \;=\; \chi^2_{S\setminus i} \;+\; \frac{(u_i - \hat{u}_i)^2}{\sigma^2_{u,i} + V_{u,i}} \;+\; \frac{(v_i - \hat{v}_i)^2}{\sigma^2_{v,i} + V_{v,i}}. \tag{9}$$

Any hit (u, v) on plane i therefore forms an accepted candidate together with the other hits of S if it lies inside the ellipse in which the last two terms add up to less than $b_i = c_S - \chi^2_{S\setminus i}$, the part of the χ^2 budget not used by the other planes. The area of this ellipse is $\pi b_i \sqrt{(\sigma^2_{u,i} + V_{u,i})(\sigma^2_{v,i} + V_{v,i})}$.

The budget b_i varies from candidate to candidate, and its mean is simple. In the whitened space of section 2.3 the accepted noise combinations fill the n_{dof}-dimensional χ^2 sphere of radius $\sqrt{c_S}$ uniformly, and eq. (9) splits the squared distance from the centre into two coordinates for plane i and $n_{\mathrm{dof}} - 2$ for the rest. For a uniformly filled sphere each coordinate carries on average $c_S/(n_{\mathrm{dof}} + 2)$, so the mean of $\chi^2_{S\setminus i}$ is $c_S(n_{\mathrm{dof}} - 2)/(n_{\mathrm{dof}} + 2)$ and

$$\langle b_i \rangle \;=\; \frac{4\, c_S}{n_{\mathrm{dof}} + 2}, \tag{10}$$

which is $c_S/2$ for $n_{\mathrm{dof}} = 6$.

The window occupancy. The expected number of *other* noise hits in the replacement window of plane i is its area times the density of the other hits,

$$\eta_i \;=\; \frac{n_i - 1}{A_i}\,\pi\,\langle b_i\rangle\,\sqrt{(\sigma_{u,i}^2 + V_{u,i})(\sigma_{v,i}^2 + V_{v,i})}, \tag{11}$$

and the *window occupancy* of the candidate is the sum over its planes,

$$\eta \;=\; \sum_{i\in S} \eta_i. \tag{12}$$

It is the expected number of other accepted candidates on the same planes that share all but one hit with the given one. In the same way, for a plane j outside S crossed by the track, the expected number of hits in the extension window is

$$\mu_j \;=\; \frac{n_j}{A_j}\,\pi\,\langle b_j\rangle\,\sqrt{(\sigma_{u,j}^2 + V_{u,j})(\sigma_{v,j}^2 + V_{v,j})}, \qquad \langle b_j\rangle = c_{S\cup j} - \frac{n_{\mathrm{dof}}(S)}{n_{\mathrm{dof}}(S)+2}\,c_S, \tag{13}$$

with $V_{u,j}$, $V_{v,j}$ now the variances of the position predicted on plane j by the fit to all of S.

Why the sets with fewest planes dominate. Every accepted candidate on $S\cup j$ contains a candidate on S, and it is accepted when a hit falls in the extension window of plane j. Therefore, approximately,

$$E_{S\cup j} \;\simeq\; \mu_j\,E_S, \tag{14}$$

for the tracks of S that cross plane j. Each additional plane multiplies the expected number by the probability that it contains a compatible hit. When the windows are nearly empty, as they must be for tracking to work at all, $\mu_j \ll 1$ and the terms with exactly k planes dominate the sum (7), by a factor $1/\mu$. In the example of section 2.6, at 10^{-3} hits/mm^2, $\mu_6 = 6.0\times10^{-4}$ for the planes 1–5 and plane 6, and eq. (14) gives $E_{123456} \simeq 6.0\times10^{-4}\times 3.2\times10^{-6} = 1.9\times10^{-9}$, as computed directly from eq. (7).

From candidates to distinct fake tracks. $E(\geq k)$ counts candidates, and the same fake can appear as several of them, in two ways. (i) Two candidates on the same planes share all but one hit (the replacement window). (ii) A candidate on $S\cup j$ contains candidates on S (the extension window). A reconstruction algorithm reports such overlapping candidates as one track. We count as one *distinct fake* each group of accepted candidates connected by sharing at least $k-1$ hits; an algorithm cannot report fewer distinct fakes than groups without losing real tracks in the same configurations.

Overlaps of type (ii) are of relative size μ, already small whenever the sets of k planes dominate. For type (i), each accepted candidate has on average η partners. Each pair is seen from both of its candidates, so when overlaps are rare the number of groups is the number of candidates minus the number of pairs, $E(\geq k)(1-\eta/2)$. We find that

$$E_{\mathrm{distinct}} \;\simeq\; e^{-\eta/2}\,E(\geq k) \tag{15}$$

describes the counted number of distinct fakes within a few per cent up to $\eta \approx 2$ (section 4), with η averaged over the accepted candidates.

Summary. The lower limit on the number of distinct fake tracks for an algorithm that accepts at least k of N planes is E_{distinct} of eq. (15), with $E(\geq k)$ from eq. (7). In the regime where tracking is possible the windows are nearly empty, $\eta \ll 1$, and the two coincide: in the example of section 2.6 the five-plane candidates have $\eta \approx 3\times10^{-3}$, so E_{distinct} and $E(\geq 5)$ differ by 0.15%. Since η grows linearly with the density, it is still only about 0.02 at the density where one fake per event is expected (7.6×10^{-3} hits/mm^2), so the correction remains small wherever the limit matters.

2.6 A numerical example

Consider six square planes at $Y = 250, 500, \ldots, 1500$ mm, forming a projective tower with an outermost plane of 1 m^2 ($W_i = Y_i/1.5$), and $\sigma = 100$ μm in both coordinates. Equation (3) gives, for six of six planes, $K = 2.0 \times 10^{-29}$ and $P = Kc^4 = 8.2 \times 10^{-24}$ at $c = 25.36$; the tower contains about 1.2×10^{15} distinguishable straight tracks. For each of the six five-plane subsets, P lies between 2.0×10^{-18} and 1.3×10^{-17}.

With a uniform noise density of 10^{-3} hits/mm^2 on every plane (from 28 hits on the smallest plane to 1000 on the largest), the expected number of fakes is 2.0×10^{-9} with six of six planes and 4.1×10^{-5} with at least five of six: the five-plane terms dominate by four orders of magnitude. Figure 3 shows how E grows with the density and with the resolution. Since the dominant terms scale as ρ^5 and σ^6, one expected fake is reached at a density that falls roughly as $\sigma^{-6/5}$: 4.0×10^{-2}, 1.7×10^{-2}, 7.6×10^{-3} and 3.3×10^{-3} hits/mm^2 for $\sigma = 25$, 50, 100 and 200 μm.

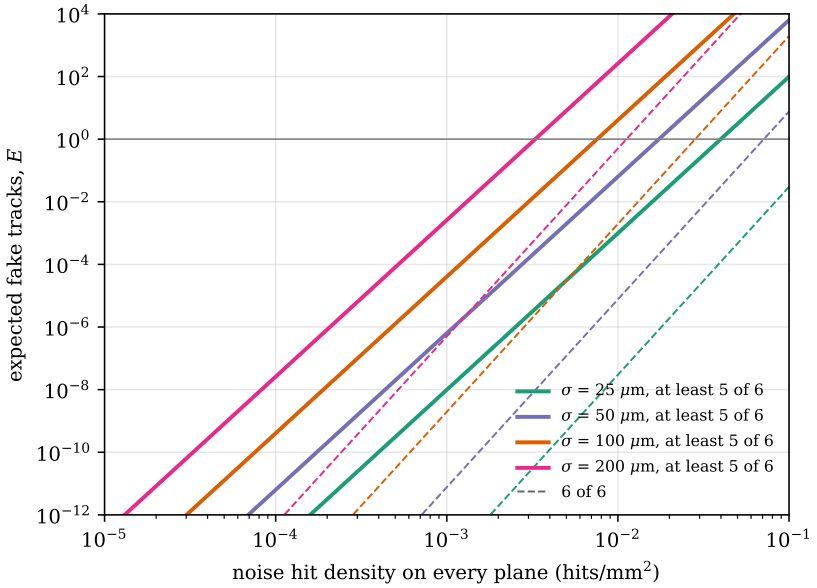


Figure 3: Expected number of fake tracks in the six-plane tower of section 2.6, against the noise hit density (the same on every plane), for four resolutions. Thick lines: at least five of six planes; dashed: six of six.

3 The general solution

3.1 Arbitrary surfaces and local densities

Nothing in the argument of section 2.3 requires the layers to be flat or parallel, or the noise to be uniform. Let each layer be a surface on which a hit measures two local coordinates, with resolutions $\sigma_{u,i}$ and $\sigma_{v,i}$, and let the noise density per unit area on layer i be $\rho_i(x)$, which may vary across the layer. A track θ crosses layer i at a point $x_i(\theta)$, with local coordinates that, divided by the resolutions, form the whitened ideal point $f(\theta)$, as before. The expected number of noise combinations on a set S of layers that pass the cut is the integral, over the positions of the hits, of the product of the densities times the indicator that the combination passes. To leading order the passing combinations again fill a tube of radius $\sqrt{c}$ around the track manifold, over which the densities are constant, and

$$E_S = V_{n_{\mathrm{dof}}}\, c^{n_{\mathrm{dof}}/2} \int_{\Theta_S} \mathrm{d}^p\theta \, \sqrt{\det\left(J_S^{\mathsf{T}} J_S\right)} \prod_{i \in S} \rho_i\big(x_i(\theta)\big)\, \sigma_{u,i}\sigma_{v,i}, \tag{16}$$

where J_S is the matrix of derivatives of the whitened local crossing coordinates on the layers of S, and Θ_S the set of accepted tracks that cross the active area of every layer of S. The volume of the box no longer appears: each density enters where the track crosses its layer. For uniform densities, $\rho_i = n_i/A_i$, eq. (16) reduces to $P_S \prod_{i\in S} n_i$ with P_S from eq. (3).

The shape and orientation of each layer enter only through J_S. A layer crossed at a grazing angle spreads the crossing points of nearby tracks over a larger area, increasing $\sqrt{\det(J_S^\mathsf{T} J_S)}$: there are more distinguishable tracks per unit of hit density. Cylinders, planes at any tilt, disks and their combinations are treated alike.

3.2 Helical tracks in a solenoidal field

In a uniform axial field we take tracks from the beam line, with parameters $\theta = (\phi_0, \kappa, z_0, t)$: initial azimuth, signed curvature, longitudinal origin and $t = \cot\vartheta$. At transverse arc length s the track is at

$$x + iy = s\,\mathrm{sinc}(\kappa s/2)\, e^{i(\phi_0+\kappa s/2)}, \qquad z = z_0 + ts, \tag{17}$$

with $\mathrm{sinc}(h) = \sin h/h$. The crossing with a coaxial cylinder of radius R has the closed form $s = (2/\kappa)\arcsin(\kappa R/2)$, at azimuth $\phi_0 + \arcsin(\kappa R/2)$; the crossing with a plane of any orientation is found by a few Newton iterations in s. The derivatives of the crossing coordinates follow analytically by implicit differentiation, so J_S costs no more than the crossing itself. The fit has $p = 4$ parameters and $n_\mathrm{dof} = 2|S|-4$. The acceptance requires $|\kappa| \le \kappa_\mathrm{max}$, a minimum transverse momentum $p_T > 0.3\,B/\kappa_\mathrm{max}$ (GeV/c, tesla, metres), and an origin within the luminous region: we assume a longitudinal beam spot of $\sigma_z = 1.5$ mm, the bunch length foreseen at the interaction point of a 10 TeV muon collider [4], and require $|z_0| < 3\sigma_z = 4.5$ mm.

The track manifold is now curved, but eq. (16) still holds to leading order; the curvature adds corrections of relative order c, as the edges do. For the fits needed in the validation we minimize χ^2 with the Levenberg–Marquardt method [26, 27], started from several values of the curvature; a single starting point would occasionally stop at a local minimum, overestimate χ^2 and underestimate the number of fakes.

Shrinking every length by λ maps a helix of curvature κ onto one of curvature $\lambda\kappa$; with $\kappa_\mathrm{max} \to \lambda\kappa_\mathrm{max}$ the family of accepted tracks is mapped onto itself and eq. (5) remains exact.

3.3 At least k of N layers in a general detector

In a real detector different tracks cross different layers: a track may cross some cylinders coaxial with the beam and some disks perpendicular to it. Equation (7) generalizes to a single integral over the track parameters, in which each track contributes the sum over the subsets of the layers it actually crosses,

$$E(\ge k) = \int \mathrm{d}^p\theta\; w(\theta), \qquad w(\theta) = \sum_{\substack{S\subseteq C(\theta)\\ |S|\ge k}} V_{n_\mathrm{dof},S}\, c_S^{\,n_\mathrm{dof},S/2} \sqrt{\det\!\left(J_S^\mathsf{T} J_S\right)} \prod_{i\in S} \rho_i\, \sigma_{u,i}\sigma_{v,i}, \tag{18}$$

where $C(\theta)$ is the set of layers crossed by track θ and each subset has its own cut. Overlapping modules within a layer, gaps and mixed combinations of cylinders and disks need no special treatment: they are all accounted for by $C(\theta)$.

3.4 Evaluating the integral

The integral of eq. (18) is evaluated by sampling track parameters uniformly over a region containing the acceptance, computing for each sampled track the layers it crosses, the derivatives J and the densities at the crossings, and averaging $w(\theta)$; the volume of the sampled region times the mean of w is E. Each sampled track carries its own contribution, $w(\theta)$, the density of fakes at

that point of track-parameter space, so the same sample gives the distribution of the expected fakes over the detector: in polar angle, along the beam, by layer combination or by region. Symmetries reduce the work: with densities independent of azimuth, ϕ_0 needs to be sampled only over a small range. The densities $\rho_i(x)$ can be taken directly from a simulation of the background, layer by layer and, where needed, as functions of position on the layer.

4 Validation

We test the results in small detectors where everything can be counted, with $\sigma = 0.1$ mm in both coordinates.

Parallel planes. Projective towers of four and six planes at $Y_i \propto i$, with sides $W_i \propto Y_i$, for straight lines (figure 4) and for helices (figure 5; in this test the depth view is written as $z = p + qr$, with r the transverse distance, which differs from the path length by less than 0.2%). For towers of growing size (smaller c') we measure P directly, by fitting random combinations until several hundred pass, and divide it by $K c'^{\,n_{\mathrm{dof}}/2}$ with K from eq. (3). The ratio tends to one as $c' \to 0$. For straight lines with four planes it is consistent with one within the statistical errors of $\pm 5\%$ for c' from 0.12 down to 0.004, where $P = 3.7 \times 10^{-7}$. For the helix with four planes it is 0.973, 0.986, 1.020 and 1.013 (± 0.02–0.03) at $c' = 0.070$, 0.031, 0.017 and 0.0077, down to $P = 3.5 \times 10^{-7}$; the six-plane points, down to $P = 4 \times 10^{-8}$, follow the same trend. At larger c' the ratio falls below one, as expected from the finite size of the sphere.

General surfaces. Four detectors of four layers each, of growing size L (figures 6 and 7): coaxial cylinders of equal length; planes tilted by 45° in the transverse plane; planes tilted by 60° towards the beam axis; and two cylinders followed by two disks perpendicular to the beam. Exact helices are fitted to random combinations of hits on the four surfaces and the passing fraction is compared with eq. (16), computed by sampling. The ratio tends to one as the detectors grow: 0.95 ± 0.05 and 1.17 ± 0.10 for the two largest sets of cylinders, 1.04 ± 0.05 and 0.99 ± 0.08 for the 45° planes, 1.09 ± 0.05 and 1.06 ± 0.09 for the 60° planes, and 0.87 ± 0.04 and 0.91 ± 0.08 for the mixed detector, where the non-projective geometry makes the finite-size corrections larger. Applied to the parallel-plane helix tower above, the same code reproduces K to 0.998 ± 0.007.

The counting identity. We generate noise-only events and fit *every* combination: one hit on each of six planes, and one hit on each plane of every five-plane subset, each with its own cut, and compare the numbers that pass with $P_S \prod_{i \in S} n_i$. For straight lines an exact pruning of the combinatorics (a partial combination already above the cut cannot pass) allows up to 1000 hits per plane and probabilities P_S down to 10^{-15}; in six configurations, with 8 to 1000 hits per plane, the ratio of counted to predicted combinations is between 0.975 and 1.037, and 0.990 ± 0.008 combined. For helices every combination is fitted by brute force, with 5 to 10 hits per plane; the ratio is 0.97 ± 0.03 over three configurations.

Distinct fakes. In the same events, candidates that share four or more hits are merged and the distinct candidates counted. For straight lines the ratio of distinct fakes to combinations follows $e^{-\eta/2}$, eq. (15), within 1–5% for $0.17 \leq \eta \leq 2.2$. For helices it falls somewhat lower at large occupancy, by a fraction of about $0.1\,\eta$, and approaches the same relation as η decreases.

Why the constant is computed, not fitted. One could instead fit K to a Monte Carlo on the reference detector, with a low-order polynomial for the finite-c' corrections. We find that such fits are biased by a few per cent for straight lines and by up to 10% for helices, and that the result depends on the fitted range and on the order of the polynomial. Equations (3) and (16)

have no such freedom; their numerical integration is precise to 10^{-4} for the towers and to better than 1% by sampling.

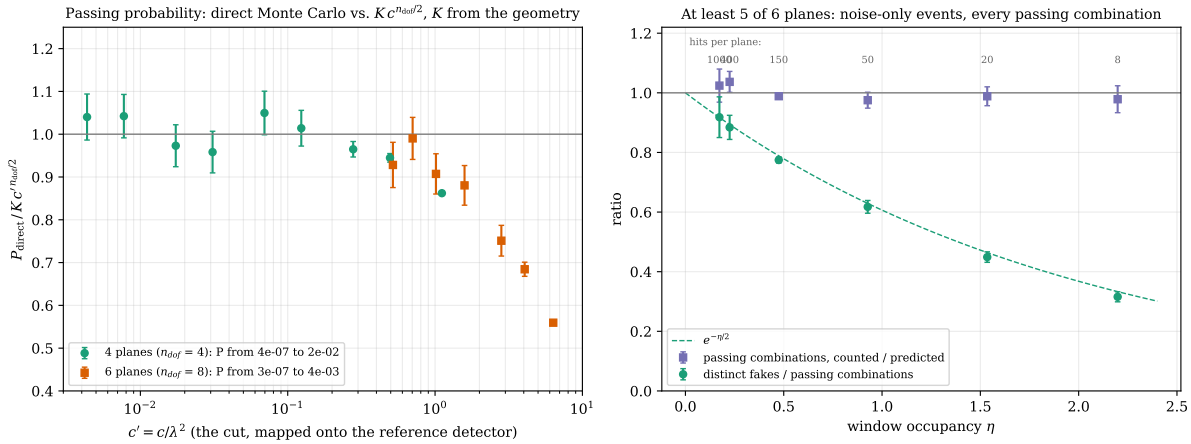


Figure 4: Parallel planes, straight lines. Left: passing probability from direct Monte Carlo divided by $Kc'^{n_{\mathrm{dof}}/2}$, with K from eq. (3), against the cut mapped onto the reference detector, for towers of four and six planes. Right: noise-only events in which every combination is fitted, for at least five of six planes; ratio of counted to predicted passing combinations, and ratio of distinct fakes to combinations, against the window occupancy η. The numbers of hits per plane are given above the points.

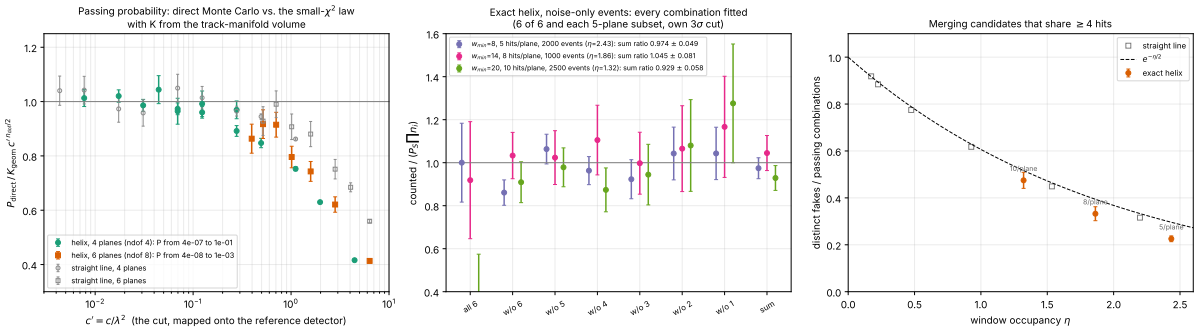


Figure 5: Parallel planes, helices. Left: as figure 4, for four and six planes, with the straight-line points in grey. Middle: counted over predicted passing combinations for six of six planes and for each five-plane subset, in three configurations. Right: distinct fakes over passing combinations against η, with the straight-line points for comparison.

5 Uncertainties and limitations

Numerical. The integrals are computed to 10^{-4} for the parallel-plane towers and to better than 1% by sampling; the finite-c corrections are negligible for detectors of realistic size. The validation tests the complete chain at the level of a few per cent, limited by Monte Carlo statistics.

Assumptions. The resolution is taken as Gaussian without tails, and multiple scattering is neglected, which is adequate for high-momentum tracks. The noise density must vary slowly on the scale of the fit windows, a few resolutions. The tracks come from the beam line; other track families (displaced tracks, cosmic rays) need their own parameters but no new formula.

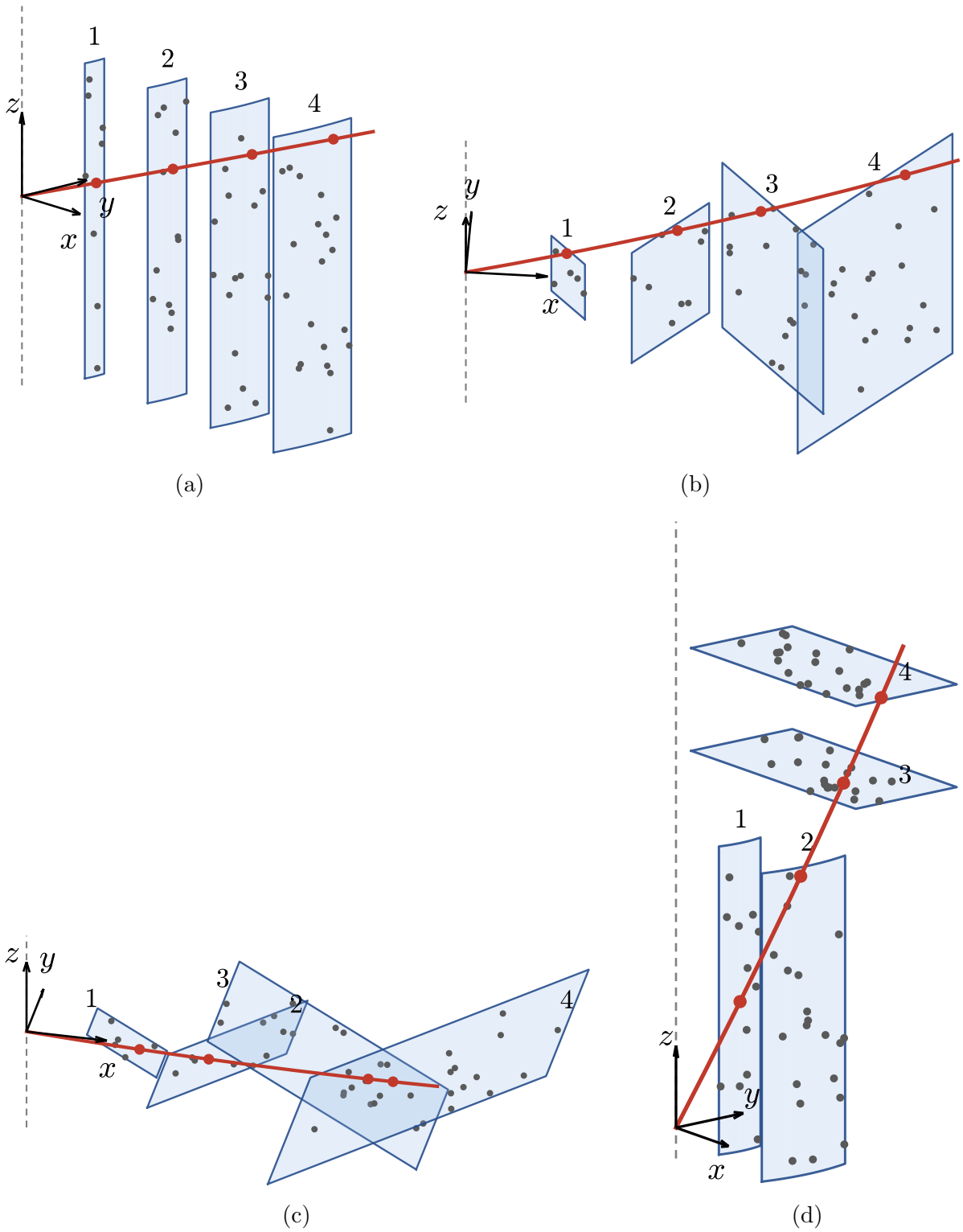


Figure 6: The four detectors used to validate the general formula, drawn to scale for one value of L: (a) four coaxial cylinders; (b) four planes tilted by 45° in the transverse plane, in alternate directions; (c) four planes tilted by 60° towards the beam axis, in alternate directions; (d) two cylinders followed by two disks perpendicular to the beam. The beam line is the z axis (dashed). Each layer carries a few uniformly distributed noise hits (grey); a helix from the beam line (red) crosses all four layers.

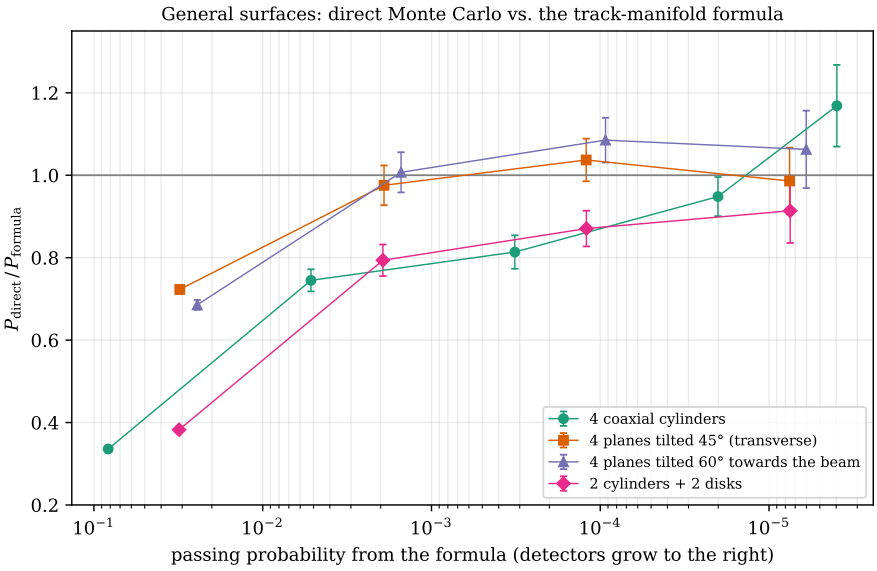


Figure 7: General surfaces, helices: passing probability from direct Monte Carlo divided by eq. (16), for four detectors of four layers each, against the probability itself (the detectors grow to the right).

What the limit does not include. Combinations mixing hits of real tracks with noise, correlated noise (clusters of hits from one particle, low-momentum particles curling in the field) and inefficiency of real tracks all add fakes or remove real tracks; none can lower the limit.

6 Conclusions

The expected number of distinct fake tracks made of noise hits that pass the goodness-of-fit cut of a track fit is a lower limit on the fake-track rate of any reconstruction algorithm that keeps the efficiency for real tracks given by that cut and acceptance. It equals the expected number of passing combinations up to a factor $e^{-\eta/2}$, where the window occupancy η is small wherever tracking is possible, and that number is the product of the noise densities and a geometric factor: the number of distinguishable tracks in the acceptance times the volume of the χ^2 sphere, everything in units of the resolution. The result holds for layers of any shape and orientation, for helical tracks in a solenoidal field and for noise densities that vary across the detector, and is evaluated by a low-dimensional integral over the track parameters, with no Monte Carlo of the rare event itself.

The limit grows as the k-th power of a common hit density for tracks with at least k of N layers, and as the n_{dof}-th power of the resolution. When hits may be missing, the sets with the fewest layers dominate, by orders of magnitude. For six planes of 1 m^2 scale and 100 μm resolution, one fake per event is reached at a noise density of about 8×10^{-3} hits/mm^2. The method gives this kind of statement for any detector from its geometry, resolution and expected occupancy alone, and can be used early in the design of a tracker to set requirements on resolution and background rejection.

A Notation

N, k	number of layers; minimum number of layers required
S, $C(\theta)$	a set of layers; the layers crossed by track θ
Y_i, W_i, A_i	depth, side and area of plane i (simple case)
$\sigma_{u,i}$, $\sigma_{v,i}$	resolutions of the two measured coordinates
n_i, ρ_i	noise hits on layer i; noise density per unit area
m, p, n_{dof}	measured coordinates, track parameters, degrees of freedom
c	cut on χ^2
Θ, Θ_B, Θ_S	acceptance; tracks inside every plane; tracks crossing every layer of S
$f(\theta)$, J	whitened ideal hits of track θ; $J = \partial f/\partial\theta$
M, $\mathrm{Vol}(M)$	track manifold; its p-dimensional volume
B, $\lvert B\rvert$	whitened box of hit positions; its volume
V_d	volume of the unit sphere in d dimensions
P, K	probability that one combination passes; $P \simeq K c^{n_{\mathrm{dof}}/2}$
E, $E(\geq k)$, $w(\theta)$	expected accepted combinations; density of fakes in track-parameter space
η	window occupancy
λ, c'	shrink factor; cut mapped onto the reference detector
ϕ_0, κ, z_0, t	helix: initial azimuth, curvature, longitudinal origin, $\cot\vartheta$

B The volume of the tube

The tube. Near the track manifold, write each point of the hit space as $x = f(\theta) + \sum_{j=1}^{n_{\mathrm{dof}}} t_j\, n_j(\theta)$, with $n_j(\theta)$ an orthonormal basis of the directions perpendicular to M at $f(\theta)$. For a point close to M, $\chi^2(x) = |t|^2$, and the fitted parameters are θ. The volume element is $\mathrm{d}^m x = \sqrt{\det(J^\mathsf{T} J)}\,(1 + O(|t|/R))\,\mathrm{d}^p\theta\,\mathrm{d}^{n_{\mathrm{dof}}}t$, with R the radius of curvature of M. Integrating t over the sphere $|t| < \sqrt{c}$ and θ over the acceptance, with the density of hit combinations evaluated at $f(\theta)$, gives eqs. (3) and (16). For a linear model $R = \infty$ and the expansion is exact away from the boundary.

The boundary. Near the edge of a layer, part of the sphere around a point of M inside the acceptance lies outside, and part of the sphere around a point just outside lies inside. For a linear model and a flat edge, a displacement t shifts the boundary of the accepted parameters linearly in t, and the average over the sphere vanishes: the two effects cancel to first order. The leading corrections come from the corners and from the curvature of M, and are of relative order c in units of the whitened layer size.

Curved layers. On a curved layer the local coordinates are coordinates on the surface, and the residuals are measured along it. When the radius of curvature of the layer is much larger than the resolution, as for any real detector, the surface is flat on the scale of the χ^2 sphere and the argument above applies unchanged.

C Constants

D Code and data availability

The code that computes the integrals for parallel planes and for general surfaces, the validation Monte Carlo and the numerical example, and the scripts that produce every number and figure of this paper, are available in a public repository [28]. [Archive the version used (e.g. Zenodo) and give its DOI.]

Table 1: The constant K of eq. (3) for the parallel-plane configurations of this paper. Validation towers: planes at $Y_i \propto i$, $W_i \propto Y_i$, $\sigma = 0.1$ mm, smallest plane 2σ wide (straight line: $Y_i = 100\,i$ mm; helix: $W_i = 0.673\,Y_i$, $\kappa_{\max} Y_N = 0.2$). Planes are numbered from the inside out.

configuration	planes	n_{dof}	K
straight line, 4 planes	all	4	1.904×10^{-2}
straight line, 6 planes	all	8	4.624×10^{-6}
	w/o 6, 5, 4, 3, 2, 1	6	$4.381, 4.149, 3.086, 1.736, 0.664, 0.701 \times 10^{-4}$
helix, 4 planes	all	4	5.833×10^{-3}
helix, 6 planes	all	8	2.043×10^{-6}
	w/o 6, 5, 4, 3, 2, 1	6	$1.385, 1.855, 1.297, 0.698, 0.297, 0.171 \times 10^{-4}$
example of section 2.6	all	8	1.988×10^{-29}

Acknowledgments

[To be written.]

Use of AI tools. [Draft, to be checked against the IOP Publishing policy at submission.] The authors used Claude (Anthropic; model Claude Opus 5.5) as an assistant to write and test the simulation and analysis code, to run the Monte Carlo validations, to search the literature, and to draft parts of the text and figures. All results were checked by the authors: the code is public [28], every number in the paper is reproduced by its scripts, and the method was validated independently as described in section 4. The authors take full responsibility for the content of this paper.

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